

GitHub Project Files Upload — Enterprise OSPF Routing Lab

One-page MNC Portfolio Guide | Repository: [Enterprise-OSPF-Routing-Lab](#)

Enterprise OSPF Routing Lab – 3 Sites (MNC Level)

HQ – Branch – Data Center | OSPF Area 0 | Dynamic Routing | End-to-End Connectivity

Why OSPF in Enterprise?

- Dynamic route learning
- Automatic failure recovery
- Scalable network design
- Fast convergence
- Supports large networks
- Industry standard protocol
- Reduces manual configuration
- Ideal for Campus, WAN & DC

Technologies Used

- Cisco IOS
- Packet Tracer
- OSPF (Area 0)
- IPv4
- Routing
- Switching
- Enterprise Networking

1 Router Configuration (OSPF)

2 IP Addressing Plan

3 OSPF Neighbor Adjacency

4 OSPF Routing Table (R3-DC)

5 End-to-End Connectivity Test

6 Troubleshooting Example

7 Key Learning Outcomes

Enterprise OSPF Network Architecture – Real World Scenario

HQ – Branch – Data Center (OSPF Area 0)

Key Highlights

- Dynamic Routing using OSPF
- Enterprise 3-Site Network
- End-to-End Connectivity
- Realistic MNC Architecture
- Unique Router IDs
- Point-to-Point WAN Links
- LAN Network Integration
- Routing Table Verification
- Network Troubleshooting
- Recursion-Ready Project

Technologies Used

- Cisco
- Packet Tracer
- OSPF (Area 0)
- IPv4
- Routing & Switching
- Enterprise Networking

2 OSPF Routing Process – How It Works (MNC Level View)

3 Network Verification & Troubleshooting – Real MNC Workflow

Enterprise OSPF Routing Lab – 3D Network Topology

HQ – Branch – Data Center (OSPF Area 0)

2 Logical Topology Diagram

3 IP Addressing Plan

4 OSPF Configuration Summary

5 Packet Flow (End-to-End Communication)

6 Cisco Packet Tracer View (2D)

7 Physical 3D Network Rack View

STEP 1 — Open Repository

Open GitHub → [Enterprise-OSPF-Routing-Lab](#).

STEP 2 — Upload Topology

Add file → Upload files → topology.png / topology.jpg.

STEP 3 — Upload Configs

Create/upload configs/router-configs.txt with R1/R2/R3 OSPF configs.

STEP 4 — Upload Screenshots

Add OSPF neighbor, routes, interface brief, protocols and ping screenshots.

STEP 5 — Upload Documentation

Create documentation/ → troubleshooting.md with the R2-R3 RCA.

STEP 6 — Upload Packet Tracer

Create lab/ → upload Enterprise-OSPF-Routing-Lab.pkt.

STEP 7 — Commit & Verify

Commit with: Initial upload - Enterprise OSPF Routing Lab; verify README + folders.

Final GitHub structure: README.md | topology/ | configs/ | screenshots/ | documentation/ | lab/ **Next:** Pin the repository on your GitHub profile, then publish the project on LinkedIn with the GitHub link.